

1

Measurement Ready

Use the tool, scale, unit, and calculation as one evidence chain.

LEARNING TARGETS

Match common lab tools to the quantity and unit they measure. • Choose an appropriate tool and justify the choice with visual evidence. • Calculate density and percent error with correct units and reasonable precision.

Three tools, three different jobs



Electronic balance

Measures mass
Typical unit: g



Graduated cylinder

Measures liquid volume
Typical unit: mL



Beaker

Holds and mixes
Volume is approximate

Measurement rules that prevent avoidable errors

- 1 Record the value and the unit together.
- 2 For an analog scale, record all certain digits plus one estimated digit.
- 3 Read a concave liquid meniscus at eye level from its lowest point.
- 4 Keep units through every calculation and round only the final result.

FORMULA TOOLBOX

density = mass / volume

percent error =
|experimental - accepted| / accepted × 100

Units tell the story

mass: g
liquid volume: mL
density: g/mL

2

Choose the Tool

Name what you see, then use visible features to justify the choice.

A. Identify each instrument

1



Electronic balance
mass
g

2



Graduated cylinder
liquid volume
mL

3



Beaker
holding/mixing; rough volume only

B. Select and justify

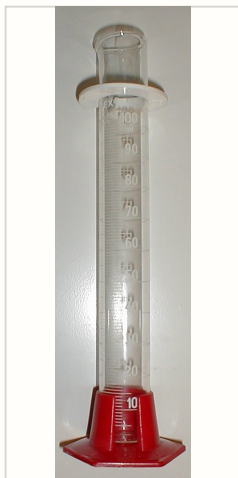
4. You need 18.6 mL of water. Choose a beaker or a graduated cylinder. Explain which visible design features support your choice.

Graduated cylinder. Its narrow shape and graduated scale support a more precise volume reading than a beaker.

5. You need about 125 mL of solution and room to stir. Choose a beaker or a graduated cylinder. Explain.

Beaker. The task asks for an approximate amount and mixing space, not a precise delivered volume.

C. Read visual evidence



6. Identify two visible features that show this vessel was designed to measure liquid volume.

Any two visible features: tall/narrow vessel, graduated marks, numbered scale, stable base, or pouring lip.

3

Calculate with Units

Show the formula, substitute values with units, and round only the final answer.

GUIDED EXAMPLE

A sample has a mass of 36.4 g and a volume of 14.0 mL.

$$d = m / V = 36.4 \text{ g} / 14.0 \text{ mL} = 2.60 \text{ g/mL}$$

7. A liquid sample has a mass of 27.0 g and a density of 2.70 g/mL. Calculate its volume.

10.0 mL

8. A liquid has a density of 0.789 g/mL and a volume of 25.0 mL. Calculate its mass.

19.7 g

9. A student measures 2.58 g/mL. The accepted value is 2.70 g/mL. Calculate percent error.

4.44%

10. Accuracy and precision

Accepted density: 2.70 g/mL

Set A 2.70, 2.71, 2.69 g/mL

Set B 2.42, 2.41, 2.42 g/mL

Which set is accurate? Which is precise? Explain using the pattern in the data.

Set A is accurate and precise. Set B is precise but not accurate.

4

Reason from Evidence

Use the data to build a claim, diagnose the error, and plan a repair.

LAB CASE: UNKNOWN LIQUID

A student uses an electronic balance and a graduated cylinder. The empty container has a mass of 42.31 g. The container plus liquid has a mass of 67.19 g. The liquid volume is 20.0 mL. The accepted density is 1.25 g/mL.

11. Mass of liquid: 24.88 g

12. Experimental density: 1.24 g/mL

13. Percent error: 0.480% (0.48% acceptable)

14. Claim - Evidence - Reasoning

Claim: Is the experimental result close to the accepted value?

Evidence: Cite at least two numbers from the case.

Reasoning: Explain what percent error tells you and what it does not tell you about precision.

The result is close to the accepted value, but the student should repeat trials before judging precision. Evidence: 1.244 g/mL before final rounding vs. 1.25 g/mL and 0.48% error.

15. Error diagnosis

Another student reads the liquid above eye level, rounds every intermediate value, and omits units. Name two problems and one exact repair action.

Read the bottom of the meniscus at eye level, record one estimated digit, keep units through every step, and repeat measurements.

REPAIR LOG

Item/skill to repair: _____

Next action: _____

Incorrect idea or step: _____

When I will retry: _____